ThreadMask: learned component rejection on a CPU baseline

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-alpha observations. ThreadMask combines a frozen disk-aware CPU segmentation baseline with learned rejection of candidate components. A fixed regularized logistic model is trained on 170 observations; one of 12 rejection thresholds is selected on 55 other observations. On a once-opened 106-observation assessment from 96 other dates, public-v6 pooled PQ increases from 0.10529 to 0.17728. The paired date-cluster bootstrap 95% improvement interval is [0.05541, 0.08763]. The filter reduces false positives while increasing false negatives. These are local results under the inspected public metric, not a leaderboard score or proof of private-scorer equivalence. The method uses no pretrained weights, test labels, paid inference or leaderboard feedback; all earlier experiments are preserved.

Final local review package, 15 October 2026. Not submitted; participant review pending.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [1]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab (www.esairlab.com/home). This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [2].

The preceding introduction retains the organizer template wording. No solution has actually been submitted. Our work is an inspectable CPU baseline, a simple learned rejection stage and an auditable evaluation workflow; Gaussian filtering, connected components and logistic regression are established methods, not claimed inventions.

2 METHODOLOGY

2.1 Frozen candidate masks

At 1024×1024, we threshold intensity at 22% of its 90th percentile, retain the largest bright connected region and fill holes to infer the solar disk. A Euclidean-distance margin rejects pixels within 24 processing pixels of its boundary. Gaussian background smoothing uses sigma 24 and divides the smoothed disk-masked intensity by smoothed disk occupancy, limiting contamination from black exterior pixels [4]. Relative darkness is background minus intensity, divided by background clipped below at one.

Candidates require relative darkness at least 0.09. One binary closing step precedes eight-connected labeling [5]. A component must contain a 0.135 strong seed, cover at least 60 processing pixels and no more than 3% of the usable disk. Labels are enlarged to the original 2048×2048 with nearest-neighbor interpolation. These frozen v2 settings are inherited without retuning for the v3 comparison.

Grayscale image → inferred disk and normalized contrast → frozen components → image-derived features → learned rejection → verified COCO RLE

Figure 1: Image-only inference pipeline. Ground truth is used for training and validation, never supplied to prediction.

2.2 Learned component rejection

For each component we extract eleven features: log area, log covariance-based elongation, bounding-box occupancy, log perimeter compactness, mean/90th-percentile/standard-deviation contrast, minimum and mean normalized disk-interior distance, log box aspect ratio and strong-seed fraction. No spine or other annotation metadata enters these features. Component geometry is never refined by the classifier: it only retains or removes an entire candidate.

Training labels are defined independently for each annotator-image entry: a candidate is positive if any ground-truth filament has IoU strictly greater than 0.5. Features are standardized using training rows only. We minimize mean binary logistic loss with fixed coefficient L2 regularization 0.01; the intercept is unpenalized. There is no class weighting, coefficient search or model-family search. Optimization converged on 5575 rows, including 657 positive and 4918 negative examples. Sigmoid scores are not separately calibration-validated.

3 EVALUATION PROTOCOL

The authorized dataset contains 707 labeled observations and 180 unlabeled test observations [2]. Earlier work inspected 270 observations across 257 dates, including v2 tuning and its 100-image comparison. Every date in every earlier split is excluded from v3 development and assessment. The earlier v2 public-v6 PQ of 0.09749 belongs to that older set and is not the baseline for the comparison below.

The remaining 437 observations span 399 dates. Before training or quality inspection, a stored SHA-256 date ordering allocates the first 96 dates to the new assessment (106 observations), the next 48 to threshold tuning (55), the next 160 to training (170), and leaves 106 observations unused. All instruments and annotators from one

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

Table 1: Same 106 observations, 96 dates and 165 annotator-image entries.

Quantity	Frozen v2	Filtered v3
PQ	0.10529	0.17728
TP	368	281
FP	2899	588
FN	802	889

date stay together. Nearby dates can still share physical structures, so date grouping does not eliminate temporal dependence.

The rejection threshold is selected from 12 prespecified values from zero to 0.9 using pooled public-v6 PQ on tuning data; ties prefer the lower threshold. Zero retains every v2 component. Threshold 0.15 achieves tuning PQ 0.14876 versus 0.08941 at zero and is frozen before the new assessment is processed. A prewritten promotion rule requires the paired bootstrap improvement interval's lower endpoint to exceed zero. This rule is met; no threshold or coefficient changes follow assessment results.

3.1 Metric and reference check

The inspected public self-evaluation notebook v6 [3] counts every GT/prediction IoU pair strictly above 0.5. Unmatched columns are false positives; unmatched rows are false negatives. TP counts, FP, FN and TP IoUs are pooled across annotator-image entries before division. This differs from the older one-to-one, image-averaged diagnostic retained in historical outputs.

On every assessment overlap matrix, both unfiltered and filtered totals exactly match the three unchanged public reference scoring functions. A NumPy adapter supplies their tensor sum interface. This checks pure arithmetic against the public code; Torch and Kaggle's private scorer were not executed. Individual sparse COCO RLE masks are also decoded and checked against exact instance pixels during export.

3.2 Matched assessment results

The v3 PQ date-cluster bootstrap 95% interval is [0.14937, 0.20403]. The paired v3-minus-v2 interval is [0.05541, 0.08763]. Both use 2000 resamples of observing dates, seed 20261014, pooling all entries within each sample. They quantify variation over these dates, not an unseen leaderboard score or uncertainty from choosing a different training dataset.

False positives fall from 2899 to 588, while true positives fall from 368 to 281 and false negatives rise from 802 to 889. The larger PQ therefore reflects a measured precision/recall tradeoff, not universally better segmentation. The filter cannot recover objects the frozen candidate generator missed and may discard thin or low-contrast true filaments.

4 RECALL-ORIENTED DEVELOPMENT AND PRECISION GUARD

After the original v3 assessment, we selected a separate recall-oriented threshold on the existing 55-image tuning set. Among seven thresholds from 0 to 0.15, we maximized recall subject to tuning PQ at least 90% of the original tuning PQ, with PQ and higher-threshold tie breaks. This selected 0.075. The model weights and candidate geometry were unchanged. On all 106 remaining unused images (95 dates), v3 gave TP=253, FP=625, FN=874 and PQ=0.16127; the recall-oriented variant gave TP=337, FP=1247, FN=790 and PQ=0.15740. Recall increased from 22.45% to 29.90%, with a paired date-bootstrap 95% gain interval of [5.59,9.47] percentage points. The corresponding PQ difference interval [-0.01527,0.00825] does not establish an improvement or equivalence. These formerly unused dates have now all been inspected.

We then prospectively declared a development search over 287 rules: no rejection, or one feature compared in either direction against 13 training quantiles for each of 11 features. A rule was eligible only if it lost no TP on either training or tuning. Of 47 eligible rules, the highest tuning PQ selected mean component distance from the disk boundary at least 0.0156783 in normalized working-resolution units. The selected rule was frozen before applying it to either assessment set; both sets were already inspected, so this is not independent validation.

On the 106-image recall assessment, this guard reduces FP from 1247 to 1094, retaining TP=337 and FN=790, and increases PQ from 0.15740 to 0.16681. On the earlier 106-image assessment, FP decreases from 1165 to 1064, retaining TP=350 and FN=820; PQ increases from 0.16564 to 0.17212. Across both inspected assessments (212 images), TP remains 687 while FP decreases from 2412 to 2158. These are pooled annotator-entry counts, not consensus physical-object counts. No entry loses a TP. This does not guarantee recall preservation on new data. Fresh image inference on three deterministic filenames agrees with the cached experiment. We retain the original v3 export as the recommended, prospectively assessed candidate; the guard remains an explicitly named development alternative, not a retroactive improvement to the original blind score.

The accompanying source, frozen rule, all search outcomes, assessment statistics and checks make the alternative reproducible. No additional candidate is submitted, no private test labels are used, and no leaderboard score is claimed.

5 REPRODUCIBILITY AND LIMITATIONS

The first three assessment filenames are illustrated in Figure 2. They are not chosen as best cases. All trained coefficients, standardization values, frozen settings, all threshold trials, per-entry statistics, data split and input hashes, source hashes and reference-parity checks accompany the code. The implementation requires pinned NumPy, SciPy, Pillow and pycocotools, and runs on CPU. Thirty-four implementation tests pass, including exact recovery of v2 with rejection disabled, geometry-preserving component removal, empty images, strict CSV fields and public-metric edge cases. A recorded pycocotools/NumPy deprecation warning does not invalidate the passed mask round trips.

The frozen v3 pipeline processes all 180 unlabeled test images and exports 950 instance rows. 11 observations have no detections; they remain explicit in the audit and contribute zero object rows, as handled by the inspected public evaluator. Every exported RLE is pixel-verified. Private upload validation remains untested. No

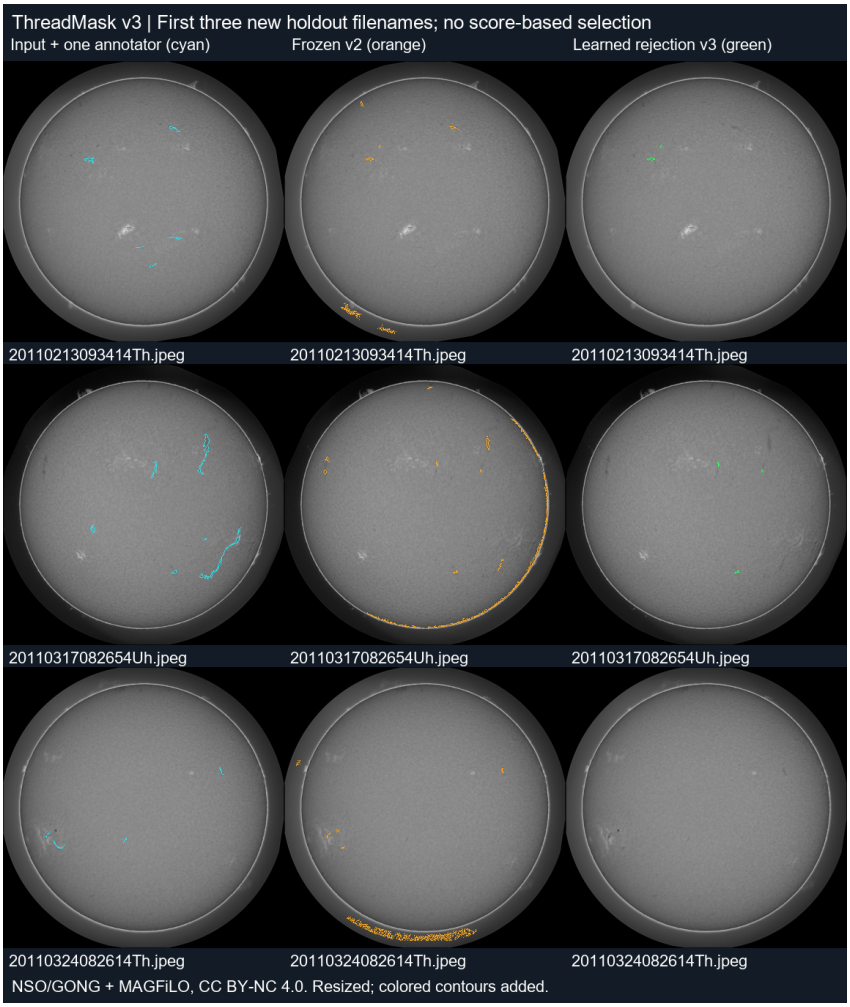


Figure 2: First three filenames in the new assessment split, without score-based selection. Cyan: one annotator; orange: v2; green: v3. NSO/GONG and MAGFiLO, CC BY-NC 4.0; resized images and colored contours.

test labels, leaderboard queries or paid compute influenced model selection.

Reproduce with `experiment_v3.py` actions freeze, fit, tune and evaluate; the stored protocol prevents silent source changes and repeated assessment overwrite. Use `predict_v3.py` for local export. The protocol, model and source hashes are checked before export. Repeating computations on this now-inspected set does not make it blind again. The earlier v1/v2 sources, measurements, CSV and report remain unchanged as historical records.

This is a small observational assessment with imperfect labels, multiple annotators and possible cross-day dependence. It does not establish competitiveness, scientific novelty, private-scorer identity or prize eligibility. A stronger evaluation now requires new observations and a separately declared untouched assessment; every existing split has been inspected. Code licensing is separate from dataset and derivative-asset terms; see the included notices. No script publishes a repository or submits a competition entry.

AI ASSISTANCE AND RELEASE STATUS

Codex assisted implementation, testing, evaluation and drafting. The participant must review, understand and reproduce the method, verify attribution and writing, and accept responsibility before any submission. The displayed author name was supplied by the participant; no expertise or organizer approval is inferred. The required subtitle, abstract opening, introductory wording and acknowledgment are retained. This local acmart document is not a submitted report; no source was edited on Overleaf.

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